1 The Price of Time

The fundamental idea is that **1 dollar today is worth more than 1 dollar in a year.** In fact, the additional time horizon will open up a broader range of investment opportunities with varying degrees of risk, including investments offering a virtually risk-free return, such as lending (under the assumption that default risk is negligible). The resulting profit is referred to as **interest**, which comes in several different forms : **simple** or **compounded**.

> **Simple interest** — interest is earned only on the initial amount invested (the principal).
>
> **Compound interest** — interest is earned on the principal *and* on the interest already accumulated.

Let r denote the simple interest rate applied to a principal amount of \$1. The compounded amount after n years is therefore given by:

$$\begin{aligned}
\$1 &\xrightarrow{1 \text{ year}} \$1 \times (1+r) = \$(1+r) \\
\$1 &\xrightarrow{2 \text{ years}} \$(1+r) \times (1+r) = \$(1+r)^2 \\
&\vdots \\
\$1 &\xrightarrow{n \text{ years}} \$(1+r)^n \qquad \text{(by simple induction)}
\end{aligned}$$

If we repeat the experiment, but now with interest compounded m times a year, that is, we receive m interest payments over one year, each at the rate r/m, then:

$$\$1 \xrightarrow{1 \text{ year}} \$\left(1+\frac{r}{m}\right)^m$$

The magic happens when we rewrite this quantity using the exponential and the natural logarithm, and let $m \to \infty$, which takes us into the framework of **continuous compounding**:

$$\begin{aligned}
\$\left(1+\frac{r}{m}\right)^m &= \$\, e^{\,m \ln\left(1+\frac{r}{m}\right)} \\
&\underset{m\to\infty}{\approx} \$\, e^{\,m \cdot \frac{r}{m}} \qquad \text{(first-order Taylor expansion, see Appendix)} \\
&= \$\, e^r
\end{aligned}$$

In summary,

$$\$1 \xrightarrow{1 \text{ year}} \$\, e^r$$

More generally, for an initial investment M_0 held for t years:

CONTINUOUS COMPOUNDING

$$M_t = M_0\, e^{rt}$$

Explore the curves interactively
Simple, discrete and continuous compounding →

Read the full proofs
Every derivation, step by step →